

Star hopping

Bright stars and distinctive constellations stand out in the night sky. Patterns such as Orion and the Big Dipper are easily found and, along with stars such as Sirius, can be used as starting points for navigating the sky, hopping from one star to the next. Some well-known star-hopping routes are shown here. These, and techniques you'll develop with practice, will help you enjoy the night sky.

Ursa Major, the great bear

The third-largest constellation, Ursa Major can be seen from anywhere in Earth's northern hemisphere. Seven of its stars make a saucepan shape called the Big Dipper; it is one of the most familiar patterns in the northern sky.



NORTHERN LATITUDES

Ursa Major and Cassiopeia are in the sky all year. They are on opposite sides of the north celestial pole and make ideal starting points for finding your way around the northern sky.

URSA MAJOR

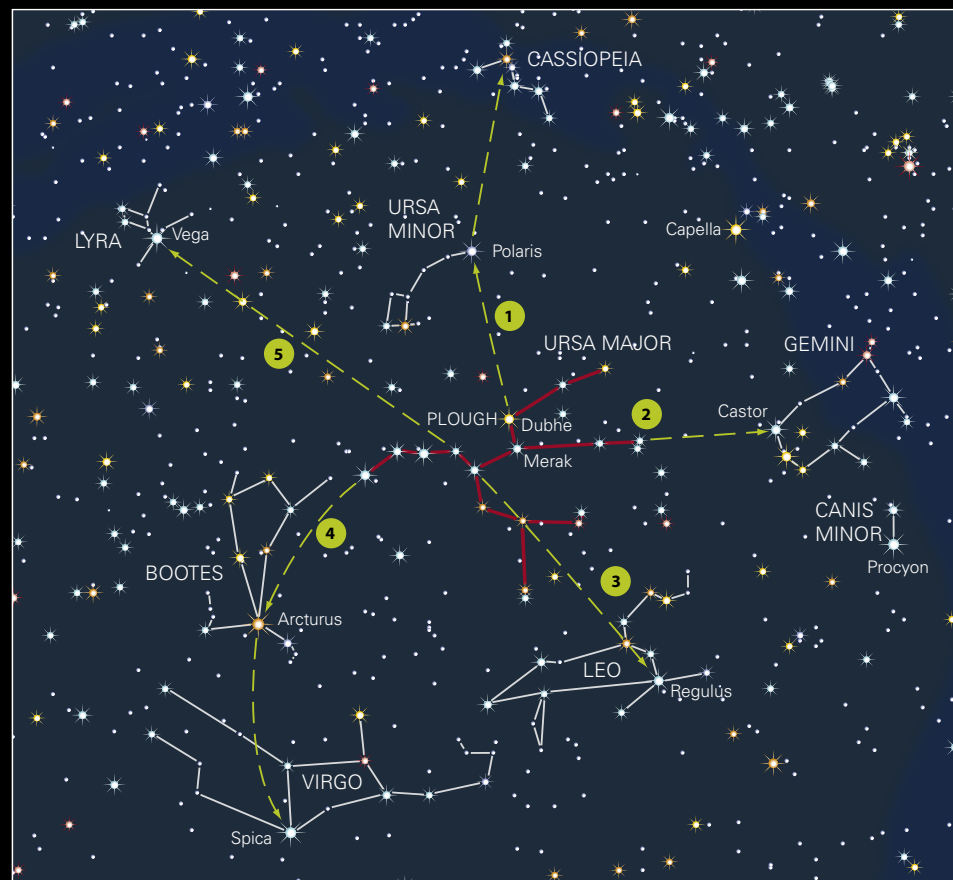
These routes all start at the Big Dipper, the simple and clear pan-shaped pattern in the tail and rump of the bear.

1 A line from Merak and Dubhe, at the right side of the pan, leads to Polaris. Also known as the Pole Star, Polaris marks the north celestial pole, around which the circumpolar stars revolve. Extend the line for the same distance again to the "W" of stars that is Cassiopeia.

2 Hop from Merak to the bear's front paw; make a second hop of a similar distance to reach Castor, the brightest star in Gemini, the twins.

3 Leo, the unambiguous shape of a crouching lion, is found by extending the line at the left side of the pan. Hop between the back legs of the bear and on to Regulus.

4 Two stars at the end of the bear's tail are the starting point to find Boötes and Virgo. Extend the line away from the bear to the brilliant star Arcturus.



URSA MAJOR

Make a second hop, the same distance again, to find Spica, the brightest star in Virgo.

5 The left end of the pan can also be used to find Lyra. Its bright star, Vega, is about the same distance away but in the opposite direction from the Big Dipper to Leo.

CASSIOPEIA

The "W" shape of Cassiopeia lies within the Milky Way and is a starting point for locating three constellations and one asterism.

1 Hop twice the length of Cassiopeia along the path of the Milky Way. The bright star you find is Deneb, representing the tail of Cygnus, the swan.

2 From Deneb, locate two brighter stars, one at either side of the Milky Way's path. These are Vega (in Lyra) and Altair (in Aquila). The three together make the Summer Triangle.



CASSIOPEIA